A REPRESENTABLE LOCALLY QUASI-FINITE MAP FROM AN AFFINE FORMAL ALGEBRAIC SPACE IS REPRESENTABLE BY SCHEMES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays one morphism between fibre products; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Affine formal algebraic spaces as filtered colimits of affine schemes, with factorisation of morphisms through a stage. Same chapter, Definition definition-affine-formal-algebraic-space, invoked at proof lines 2390-2392. Supplies the factorisation of the restriction to a quasi-compact open subspace.
- Local quasi-finiteness is preserved by base change. Morphisms of Spaces, Lemma spaces-morphisms-lemma-permanence-quasi-finite, invoked at proof lines 2399-2402. Gives local quasi-finiteness of the morphism from the quasi-compact open subspace.
- A locally quasi-finite separated algebraic space over a scheme is a scheme, and a criterion for being a scheme via quasi-compact open subspaces. Morphisms of Spaces, Proposition spaces-morphisms-proposition-locally-quasi-finite-separated-over-scheme at proof lines 2402-2405, and Properties of Spaces, Lemma spaces-properties-lemma-subscheme at proof lines 2405-2407. Make each quasi-compact open subspace a scheme and then the whole fibre product a scheme.

Source reference. Original lemma 03XN: sources/stacks-chow/tags.txt, source lines 5154–5154. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original proposition 03XX: sources/stacks-chow/tags.txt, source lines 5164–5164. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03JH: sources/stacks-chow/tags.txt, source lines 4639–4639. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is an *affine formal algebraic space* if there exist

(1) a directed set Λ,
(2) a system $(X_\lambda, f_{\lambda\mu})$ over Λ in $(Sch/S)_{fppf}$ where
   (a) each X_λ is affine,
   (b) each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a thickening,

such that

$$X \cong \operatorname{colim}_{\lambda\in\Lambda} X_\lambda$$

as fppf sheaves and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of affine formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a map of presheaves on $(Sch/S)_{fppf}$. If X is an affine formal algebraic space and f is representable by algebraic spaces and locally quasi-finite, then f is representable (by schemes).*

Proof. Let T be a scheme over S and $T \to Y$ a map. We have to show that the algebraic space $X \times_Y T$ is a scheme. Write $X = \operatorname{colim} X_\lambda$ as in Definition 0.1. Let $W \subset X \times_Y T$ be a quasi-compact open subspace. The restriction of the projection $X \times_Y T \to X$ to W factors through X_λ for some λ. Then

$$W \to X_\lambda \times_S T$$

is a monomorphism (hence separated) and locally quasi-finite (because $W \to X \times_Y T \to T$ is locally quasi-finite by our assumption on $X \to Y$, see Morphisms of Spaces, Lemma 03XN). Hence W is a scheme by Morphisms of Spaces, Proposition 03XX. Thus $X \times_Y T$ is a scheme by Properties of Spaces, Lemma 03JH. □